MALAYSIA PRICE WATCH

Grocery Price Intelligence Report

Prepared for: KPDN price monitoring, a supermarket category team and an FMCG procurement team | Data: Oct 2025 to Sep 2026 | Built with PostgreSQL, SQL, Python and Power BI | Moztoba Mutashim Billah, October 2026

Executive summary

The business problem. Malaysian households, retailers and suppliers make pricing and buying decisions with national averages, which hide where and in which kind of shop the same item costs more. This report turns 21 million government shelf-price records into a KPI scorecard for three stakeholders, the insights behind it, and an action plan.

9.9%
more for the same basket in Kuala Lumpur than in Kedah (RM 390 on 265 items)

12.0%
more for 8 everyday staples at wet markets than at hypermarkets

99.95%
of shops kept sugar and cooking oil at or below the controlled price

58%
of all price increases came from 3 fresh categories holding 17% of items

12.1 pts
gap between the average supermarket and the cheapest chain

December
the high-risk month: 78 price-spike weeks, 74 in fresh produce

Table 1 **Key findings and what they mean for the business**
One line per finding: the evidence, the business impact and who should act.

Finding	Business impact	Owner
Location and shop format move prices by similar amounts: 9.9% between states, 9.0% between hypermarkets and sundry shops on identical items.	Monitoring and pricing set by state average miss the shop-format effect, which is as large as the location effect.	KPDN, retailer
Price controls work: sugar and cooking oil sat at the controlled price in 99.95% of shop-months. The gaps are in uncontrolled fresh produce and dry goods.	Enforcement effort can move from controlled items to fresh produce in the dearest districts.	KPDN
Wet markets are not cheapest for fresh food (index 106.5 against 97.7 at hypermarkets); large formats win in 14 of 16 states.	Fresh produce is where a supermarket can win on price perception; Kuala Lumpur supermarkets are the exception at 115.8.	Retailer
Cost risk is concentrated: 3 fresh categories caused 58% of increases, and fresh prices swing 8 times more than packaged goods.	Fixed contracts suit packaged goods; vegetables, onions and meat need indexed contracts and a second source.	Procurement
No festive price hike: festive staples fell 2.4% before Chinese New Year and 5.0% before Hari Raya Aidilfitri. December, not festivals, had the most spikes.	Lock fresh-staple prices before December; festive stock builds carry less price risk than assumed.	Procurement, KPDN

1 Business context and KPI framework

Each stakeholder faces a different decision, so each gets its own KPIs (Table 2). Every KPI is calculated in SQL from the same cleaned database and is refreshed by rerunning one script.

Table 2 Stakeholders and decisions

Who uses the report and for which decision; their KPIs follow in Table 3.

Stakeholder	Business problem	Decision supported
KPDN price monitoring	Limited inspectors, nationwide complaints	Where to monitor and enforce
Supermarket category manager	Shoppers compare formats and chains	Which categories to price against whom
FMCG procurement analyst	Input costs move without warning	Contract type and timing of purchases

Table 3 KPI scorecard

Values for September 2026 or the full year as stated. Alert rules are proposed thresholds for a monthly review; status follows the rule.

KPI	Definition	Value	Alert rule	Status
KPDN price monitoring				
Regional basket gap	Same 265-item basket, dearest vs cheapest state	9.9% (RM 390)	> 5%	**Act**
District price spread	Widest district gap inside one state, index points	11.7 (KL)	> 8	**Act**
Monitoring coverage	Districts with 4+ reporting shops	76.4%	< 90%	**Watch**
Price-control adherence	Shop-months at or below the controlled price, sugar and cooking oil	99.95%	< 99%	**On track**
Supermarket chain, category manager				
Price position index	Average supermarket vs national median (100)	101.4	> 100	**Watch**
Gap to price leader	Index points behind the cheapest chain (Tesco, 89.3)	12.1	> 5	**Act**
Items priced above market	Supermarket items above the national median	32.4%	> 25%	**Watch**
Fresh produce index	Supermarket fresh produce vs national median	100.9; KL 115.8	> 105	**Act (KL)**
FMCG procurement				
Annual price change	Median item, same shops, Oct 2025 to Sep 2026	0.0%	> 3%	**On track**
Increase concentration	Share of all increases from the top 3 categories	58%	> 50%	**Act**
Fresh price volatility	Coefficient of variation of monthly prices (packaged: 0.87%)	6.92%	> 5%	**Act**
Spike frequency	Item-weeks 2 standard deviations above the prior 8 weeks	4.2%; Dec 78 weeks	Dec peak	**Watch**

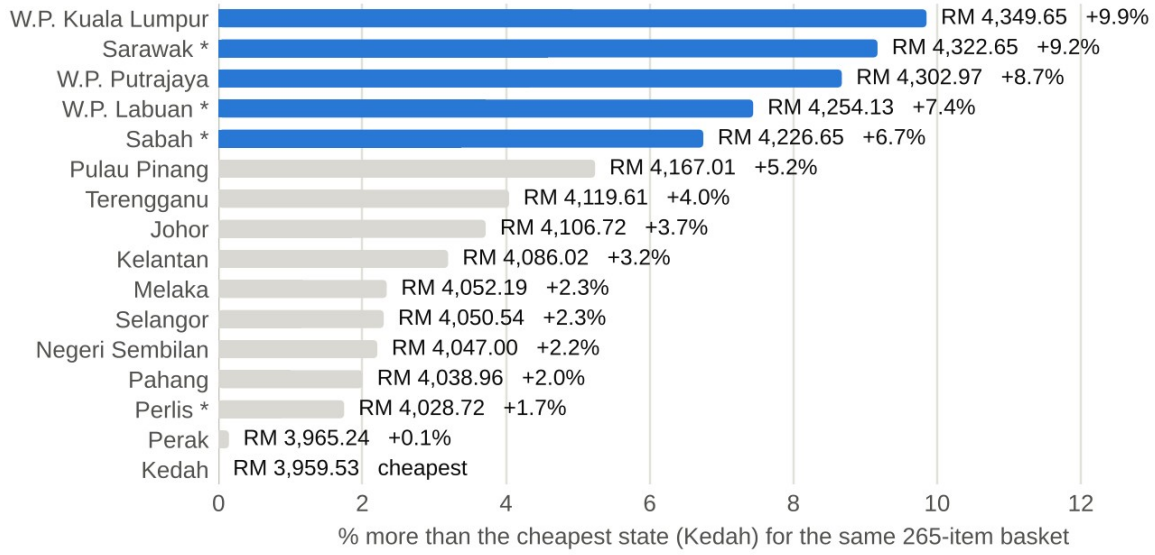
Sources: queries q07, q09, q10, q11, q12, q13, q14, q23, q29 and q31 (Appendix A).

2 KPDN: affordability and monitoring

Where households pay more. The same 265-item basket cost RM 4,349.65 in Kuala Lumpur and RM 3,959.53 in Kedah, RM 390.12 or 9.9% more (Figure 1). The five dearest are the three federal territories, Sabah and Sarawak.

Figure 1 The same basket cost up to 9.9% more depending on the state

265-item basket at each state's median prices, September 2026. * Ten or more items not priced in the state, filled with the national median.

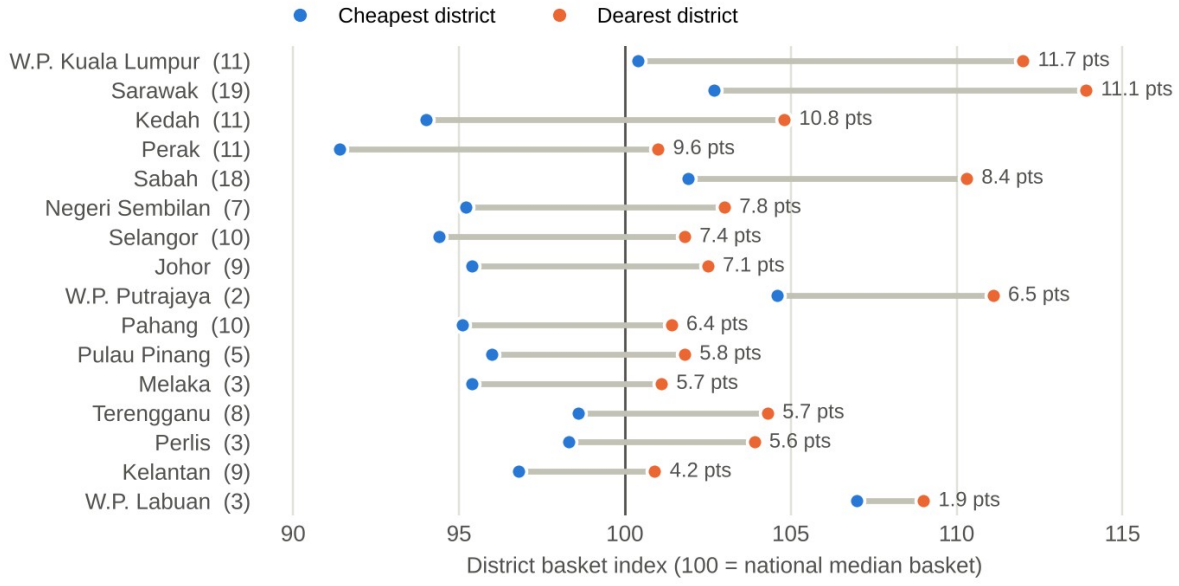


Source: query q09.

State averages hide the hot spots. Inside Kuala Lumpur, district basket indexes run from 100.4 (Bandar Tun Razak) to 112.0 (Segambut); five states have a district spread above 8 points (Figure 2). The dearest districts nationally are Limbang (113.9), Segambut (112.0) and Lawas (111.4).

Figure 2 Districts in the same state differ by up to 11.7 index points

Cheapest and dearest district basket index in each state, September 2026 (100 = national median basket).



Source: query q10. Gaps are calculated before rounding.

Price controls are working; the gaps are elsewhere. Coarse sugar (ceiling RM 2.85) and packet cooking oil (RM 2.50) were at or below the controlled price in 99.95% of 34,609 shop-months (Bernama, 2024; Malay Mail, 2026). The price gaps in this report come from uncontrolled fresh produce and dry goods, so that is where inspection time earns the most.

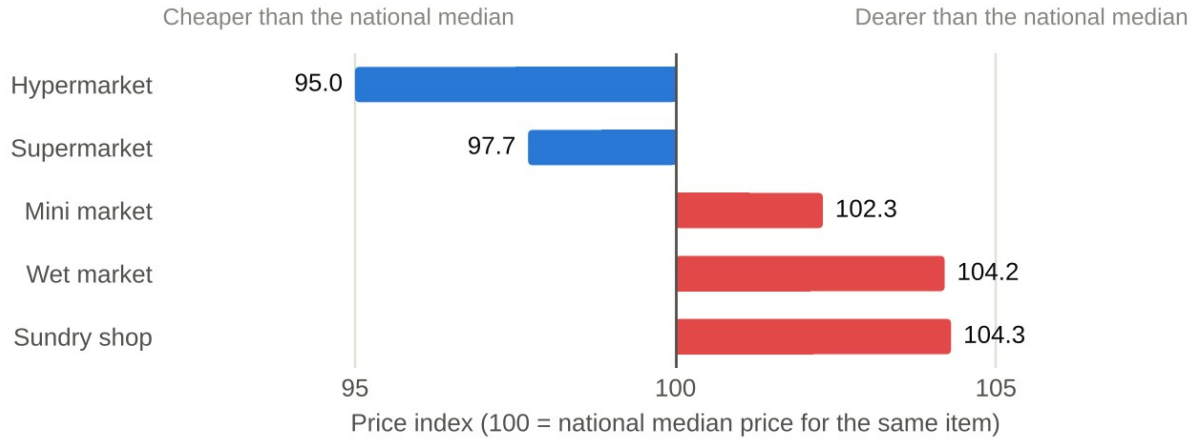
Coverage is thinnest where prices are highest. Only 76.4% of listed districts had four or more reporting shops in September 2026; 22 of the 30 districts with three or fewer are in Sabah or Sarawak.

> **What KPDN should do.** Shift inspections from controlled items to fresh produce and dry goods in the dearest districts; add reporting shops in the thin Sabah and Sarawak districts; run a weekly vegetable price watch from November to January.

3 Supermarket chain: price positioning

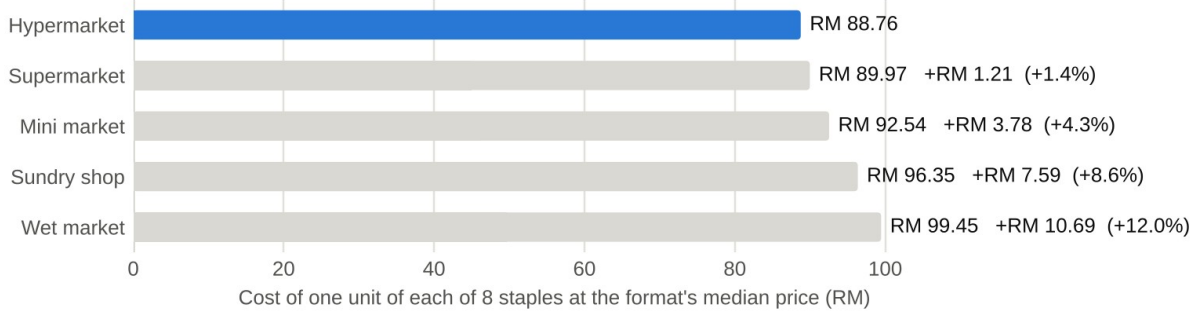
Shoppers pay most at small formats. On 181 items sold by all five formats, hypermarkets priced 5.0% below the national median and sundry shops 4.3% above it (Figure 3). For eight everyday staples, one unit of each cost RM 88.76 at hypermarkets and RM 99.45 at wet markets (Figure 4).

Figure 3 **Hypermarkets were the cheapest format on identical items**
Price index by shop type, 181 items sold in all five formats, September 2026 (100 = national median).



Source: query q03.

Figure 4 **Eight staples cost 12.0% more at wet markets than at hypermarkets**
Chicken, eggs, onions, garlic, red chilli, rice, sugar and cooking oil; median prices, September 2026.

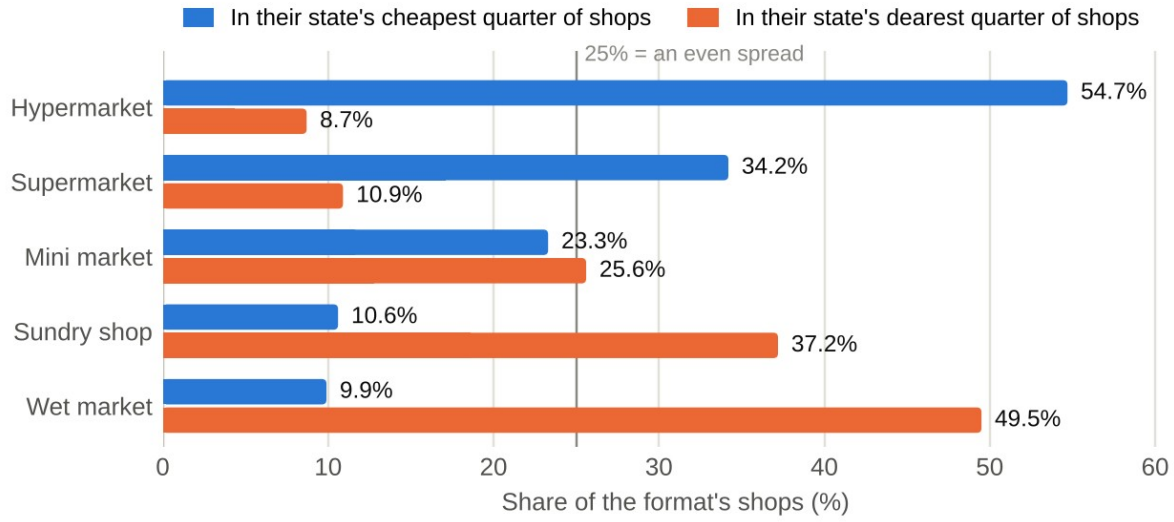


Source: query q03.

Supermarkets sit in the middle, 12.1 points behind the price leaders. The average supermarket scored 101.4, against 89.3 for Tesco, 89.4 for Lotus's and 90.2 for Econsave's 58

branches. Hypermarkets cluster at the cheap end of their state, with 54.7% in the cheapest quarter of shops; half of wet markets (49.5%) sit in the dearest quarter (Figure 5).

Figure 5 Hypermarkets cluster at the cheap end of their state, wet markets at the dear end
Share of each format's shops in the cheapest and dearest quarter of shops in their state, September 2026.



Source: queries q11 and q19.

Where to compete. Branded packaged goods sit within a few points of 100 in every format, so there is little to win there. The gaps are in fresh produce and dry goods (Table 4). Wet markets, assumed to be the cheap option for fresh food, averaged 106.5 against 97.7 at hypermarkets and lost to a large format in 14 of 16 states. The exception that matters for a supermarket chain is Kuala Lumpur, where supermarket fresh produce scored 115.8 against 100.8 at hypermarkets.

Table 4 Price index by product group and format, September 2026
Shops pricing 30 or more items. Cell colour: blue below the national median (100), red above it; lighter to darker at 1, 5 and 10 points from 100.

Product group	Items	Hyper-market	Super-market	Mini market	Wet market	Sundry shop
Fresh produce	115	97.7	100.9	109.4	106.5	113.0
Dry goods	22	94.4	99.6	107.1	119.4	112.8
Packaged goods	109	98.2	100.6	100.2	104.0	102.7
Milk and baby products	25	99.2	101.8	100.7	104.3	105.3
Household and hygiene	27	97.5	100.0	100.3	102.9	106.1
Beverages	31	98.6	101.3	99.9	101.1	103.1

Source: query q12.

> **What the category manager should do.** Benchmark fresh produce and dry goods against Tesco, Lotus's and Econsave rather than independent shops; review Kuala Lumpur fresh produce pricing first; track the gap to the price leader monthly.

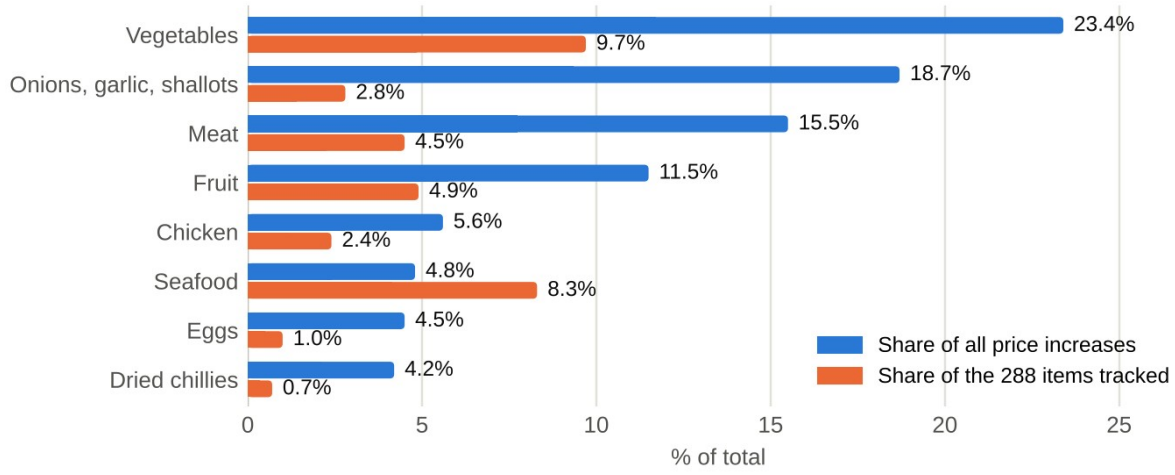
4 FMCG procurement: cost risk

Most prices were flat; the risk sits in a few categories. In the same shops, 84% of 288 items moved less than 5% over the year and the median item did not change. Vegetables, onions and

garlic, and meat produced 58% of all increases with 17% of the items (Figure 6). Tomatoes rose 69.4%.

Figure 6 Three fresh categories produced most of the year's increases

Share of all price increases and share of the 288 items, by category, same shops, October 2025 to September 2026.

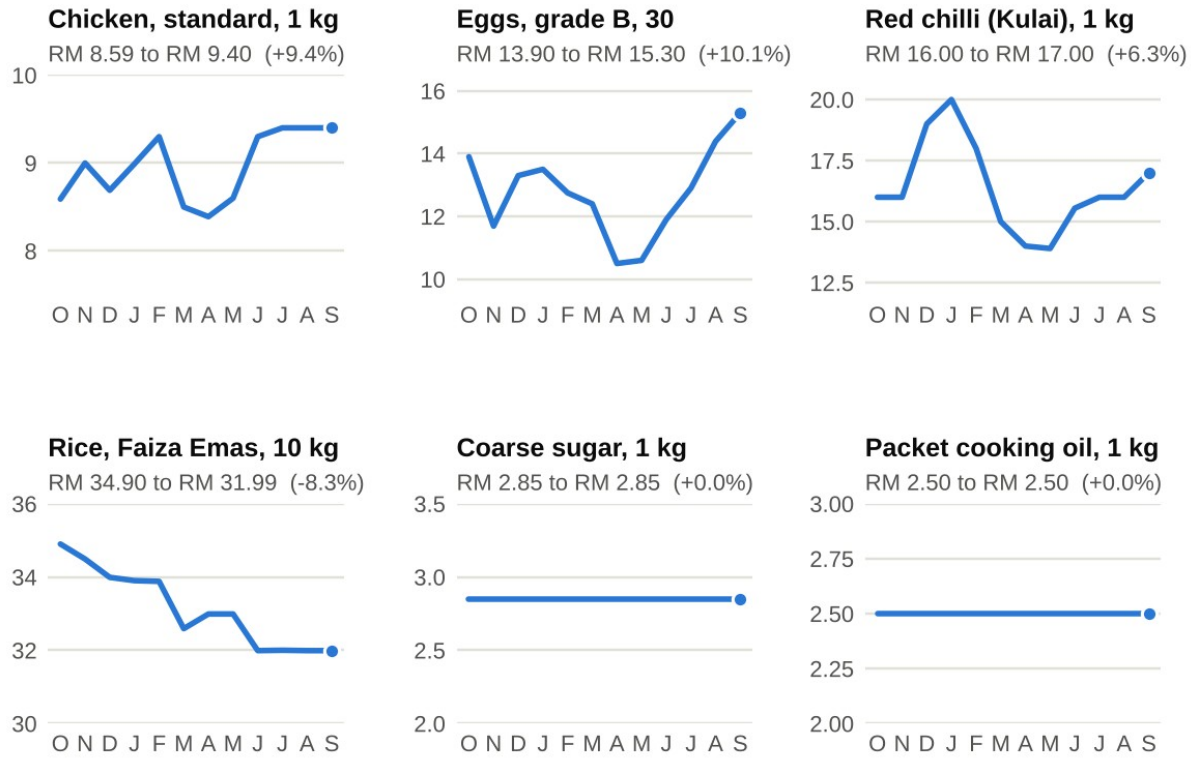


Source: query q14.

Volatility, not the annual change, is the budget risk. Fresh produce prices varied 6.92% month to month on average against 0.87% for packaged goods. Red chilli (cili akar) ranged from RM 13.99 to RM 32.00 a kilogram across the year's monthly medians, 2.3 times from low to high (Table 5). Price-controlled sugar and cooking oil never moved (Figure 7).

Figure 7 Controlled staples stood still; chicken, eggs and chilli did not

National median price by month, October 2025 to September 2026.



Source: queries q17 and q20.

Table 5 **The most volatile items**

Lowest and highest monthly national median and the coefficient of variation, October 2025 to September 2026.

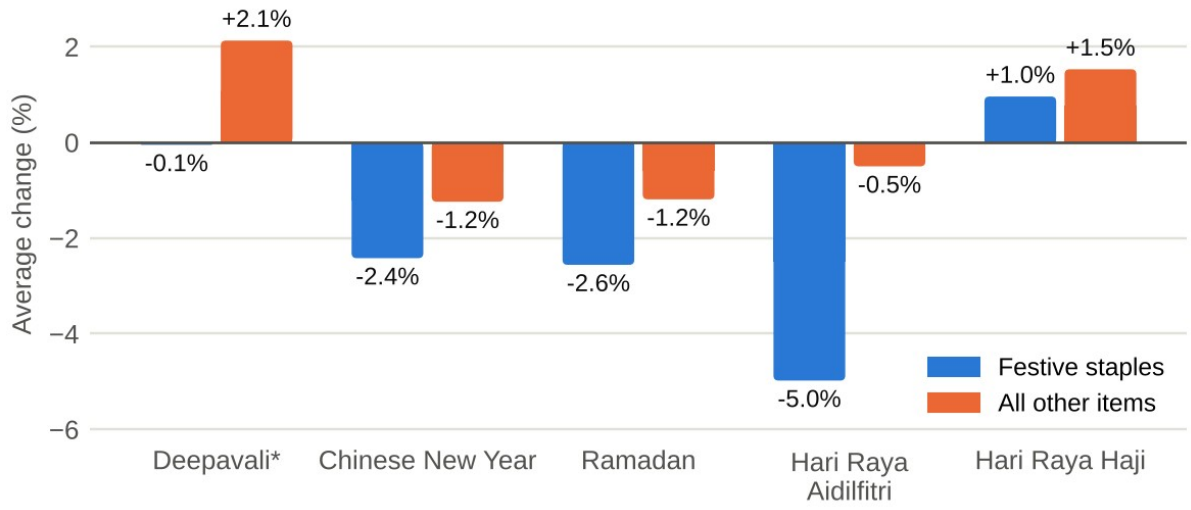
Item (1 kg)	Lowest (RM)	Highest (RM)	Variation
Red chilli (cili akar)	13.99	32.00	33.4%
Shallots, rose variety, imported (India)	5.70	12.99	32.3%
Red grapes, seeded	10.50	22.90	28.3%
Tomato	4.00	9.49	26.8%
Seedless red watermelon	1.99	4.49	22.1%

Source: query q29.

The risk window is December, not the festive season. Festive staples fell before Chinese New Year (−2.4%) and Hari Raya Aidilfitri (−5.0%) in the same shops (Figure 8). Price spikes were less frequent before festivals (2.83% of item-weeks) than in other weeks (4.52%), and December 2025 had 78 spike weeks, 74 of them fresh produce (Figure 9).

Figure 8 **Festive staples got cheaper, not dearer, before Chinese New Year and Hari Raya Aidilfitri**

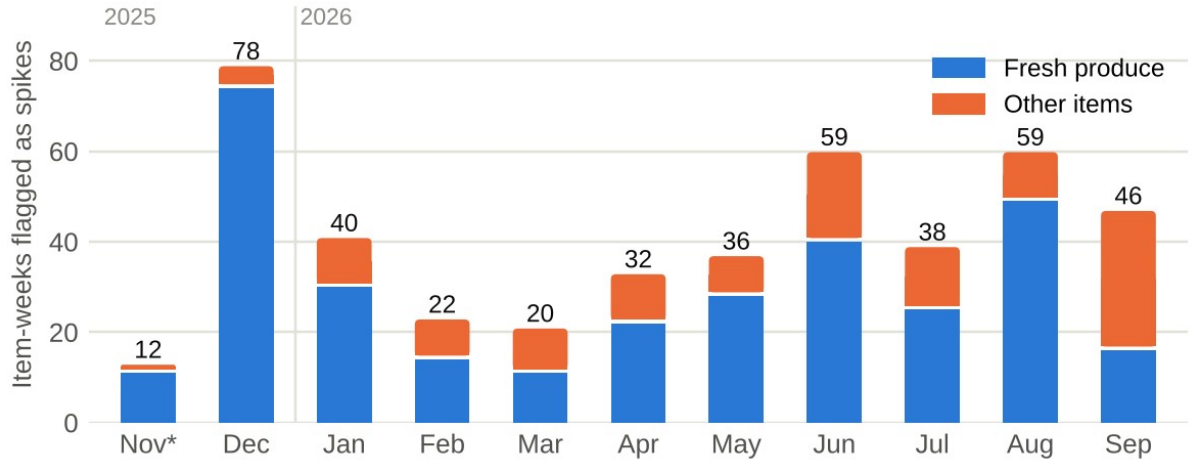
Average change in the same shops: 14 days before each festival against days 42 to 15 before it. * Deepavali baseline covers only 1 to 5 October.



Source: query q21.

Figure 9 **December had the most price spikes**

Item-weeks flagged as price spikes, by month. * November 2025 is partial.



Source: query q23.

What procurement should do. Lock fixed-price contracts for packaged goods; use indexed contracts and a second source for vegetables, onions and meat; lock fresh-staple prices in October and November, ahead of the December spike window.

5 Action plan

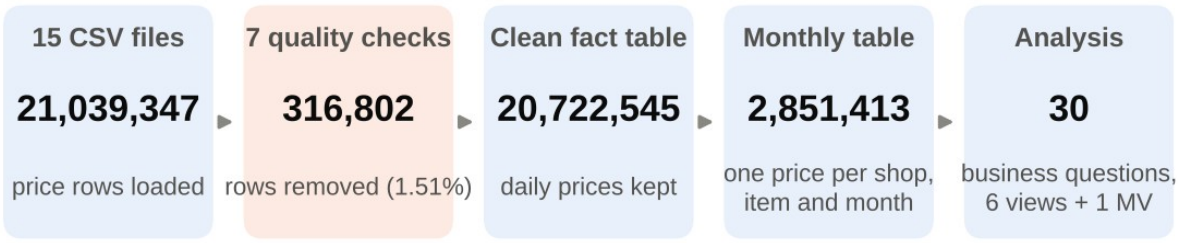
Table 6 **Recommended actions**
Each action, its owner, the KPI that tracks it and a proposed target.

#	Action	Owner	KPI tracked	Proposed target	When
1	Move inspections from controlled items to fresh produce and dry goods in the ten dearest districts	KPDN enforcement	District price spread	Below 8 points	Next quarter
2	Add reporting shops in the 22 Sabah and Sarawak districts with three or fewer	KPDN PriceCatcher team	Monitoring coverage	90% of districts	Within 6 months
3	Run a weekly vegetable price watch from November to January	KPDN, procurement	Spike frequency	Alerts within one week	Every November
4	Benchmark fresh produce and dry goods against Tesco, Lotus's and Econsave	Category manager	Gap to price leader	Below 5 points	Monthly
5	Review Kuala Lumpur supermarket fresh produce pricing	Category manager, KL	Fresh produce index (KL)	Below 105	Within 3 months
6	Fixed contracts for packaged goods; indexed contracts and a second source for vegetables, onions and meat	Procurement	Fresh price volatility	Budget variance within 5%	Next contract cycle
7	Lock fresh-staple prices with suppliers in October and November	Procurement	Spike frequency	No December buying at spike prices	Yearly

The KPIs above are built to refresh monthly: new PriceCatcher files are loaded, the build script reruns, and the reporting views feed a three-page Power BI dashboard, one page per stakeholder.

6 How the analysis was built

Figure 10 **From 21 million raw rows to a refreshable KPI scorecard**
Row counts at each stage of the build.



Source: build run log.

Table 7 Solution summary
What was done at each stage and the evidence of quality.

Stage	What was done	Result
Data	12 monthly PriceCatcher files, item and shop lookups, DOSM CPI by state	21,039,347 price records, 2,262 shops, 338 items, 16 states
Data quality	7 documented checks: impossible prices, duplicates, unknown items and shops, outliers, shop churn	1.51% removed; 99% of that from 17 item codes missing in the published list
Data model	PostgreSQL star schema with a daily fact table and a monthly one-price-per-shop table	Each shop counts once in every comparison
Analysis	30 stakeholder questions in SQL: CTEs, window functions, GROUPING SETS, matched-shop comparisons	The KPIs and findings in this report
Validation	PriceCatcher food prices compared with DOSM's food CPI by state	Correlation 0.60 over 160 state-months
Performance	Composite index; materialized view for the weekly spike table	2.97 s to 0.09 s; 52.5 s to 0.05 s
Delivery	6 reporting views plus 1 materialized view for Power BI; full rebuild script	Refreshable monthly in about 13 minutes

Three method choices make the numbers trustworthy. Wet markets report each item about 15 times a month and sundry shops about 3, so every comparison uses one price per shop. Shops join and leave during the year, so changes over time compare the same shops: on a naive comparison, imported Pakistani onions rose 3.7% instead of 35.2%. Most small shops are surveyed on Mondays, so spikes are measured weekly, not daily.

7 Limitations

- PriceCatcher is collected for price surveillance; inflation is measured by DOSM's CPI (Ministry of Domestic Trade and Cost of Living & Department of Statistics Malaysia, 2026), used here only as a cross-check.
- Every item has equal weight, so baskets compare like-for-like prices rather than household spending.
- Format and location comparisons use September 2026 (the queries run for any month); freshness and bargaining are not recorded.

References

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Appendix A The 30 SQL analyses behind the KPIs

Each analysis is one SQL file in the project repository, written for one stakeholder: A KPDN, B category manager, C procurement. q31 adds the price-control adherence and monitoring coverage KPIs.

Table 8 Analysis index

What each query answers and the SQL techniques it uses.

#	For	Business question	SQL techniques
q01	A	National median price of each item, September 2026	JOIN, WHERE, GROUP BY, PERCENTILE_CONT ... WITHIN GROUP
q02	A	Most and least expensive state for the 10 most common items	CTEs, JOIN, AVG, ORDER BY, LIMIT, RANK()
q03	B	How prices differ by premise type for the same item	GROUP BY two columns, HAVING, FILTER inside an aggregate
q04	B	Items with the biggest gap between the cheapest and dearest shop type	CTE, MAX − MIN, ARRAY_AGG
q05	C	How many items and premises does each state have? Is coverage balanced?	COUNT(DISTINCT), share of total with SUM() OVER ()
q06	A	Districts with the fewest reporting premises	GROUP BY, HAVING (filters groups, after counting)
q07	B	How many items does each shop type price above the national median?	CASE inside SUM, the FILTER clause
q08	C	Which item categories have the widest price range across Malaysia?	STDDEV, AVG, PERCENTILE_CONT at 0.1 and 0.9
q09	A	What does an "essentials basket" cost in each state?	multi-step CTE, CROSS JOIN, LEFT JOIN, COALESCE
q10	A	Which district has the cheapest basket within each state?	CTE + correlated subquery in WHERE, TEMP TABLE
q11	B	Premise price index: which shops are cheapest overall?	CTE, ratio metric, HAVING
q12	B	Premise price index by premise type and state	CTE + GROUP BY, FILTER to pivot types into columns
q13	C	Which items changed price most between October 2025 and September 2026?	CTE with two periods, JOIN on two keys, % change
q14	C	Are price increases concentrated in a few categories?	CTE, share of total with SUM() OVER (), running share
q15	B	Rank premises by price for each item within each district	RANK() OVER (PARTITION BY ...)
q16	B	For each item and district, which premise is cheapest most often?	DENSE_RANK + COUNT, ROW_NUMBER to keep the top row per group
q17	C	Month-on-month median price change for each item	LAG() OVER (PARTITION BY ... ORDER BY ...)
q18	C	Rolling 4-week average price for five key staples	AVG() OVER (ROWS BETWEEN 3 PRECEDING AND CURRENT ROW)
q19	A	Split premises into price quartiles within each state	NTILE(4)
q20	C	Running cumulative price change since October 2025, per item	SUM() OVER (PARTITION BY ... ORDER BY ...) = running total
q21	C	Do festive staples get dearer in the 2 weeks before each festival?	CTE + date windows built from a festival table, CROSS JOIN
q22	A	Which states see the biggest festive price spikes?	window function (RANK) over a GROUP BY result

#	For	Business question	SQL techniques
q23	C	Detect price spikes: weeks far above an item's recent normal	AVG() and STDDEV() OVER (ROWS BETWEEN 8 PRECEDING AND 1 PRECEDING)
q24	B	How often does each premise change its price on the same item?	LAG + COUNT of changes, then averages by group
q25	B	First and latest price per premise and item over the year	FIRST_VALUE / LAST_VALUE, window frames, ROW_NUMBER to keep one row per group
q26	A	One summary table: by state, by premise type, and grand total	GROUPING SETS, GROUPING()
q27	All	Sense check: do PriceCatcher food prices move with CPI food (division 01)?	JOIN to a reference table, LN/EXP, CORR()
q28	B	Premises that stopped reporting for a month or more	generate_series + anti-join, LATERAL
q29	C	Which items have the most stable prices?	STDDEV ÷ AVG, HAVING on a count of months
q30	All	The three most important findings, as one small report table	chained CTEs, UNION ALL, string building with || and TO_CHAR
q31	A	KPI checks: price-control adherence and monitoring coverage	CASE inside AVG, UNION ALL, CTEs, COUNT(DISTINCT)